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This document is identical to document 3210-8440-SP-0013 Rev A1 with the sole exception of Document number and Document number cross references

T1 revision amendments:
1. Wherever Paradip Refinery Project is specified it shall be read as 'Ethylene Glycol Project (IEGP)'.
2. Wherever Approval / agreement /review requirement by IOCL is specified, it shall be read as 'by PMC/ IOCL'.

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1.0 SCOPE

This Engineering Standard covers the noise limits for equipment and specifies the information which shall be submitted by the equipment vendor with his quotation. Any deviations from this Standard may only be made with the prior written approval of the IOCL.

2.0 SPECIFICATIONS AND STANDARDS

2.1 International Standard

Reference is made to the following EEMUA (Engineering Equipment and Material Users Association) publications which to the extent indicated, form part of this Specification. The latest revision of the applicable documents shall apply:

104 Noise: A guide to Information required from Equipment Vendors.

106 General Specification for Silencers and Acoustic Enclosures.

140 Noise Procedure Specification.

141 Guide to the Use of Noise Procedure Specification.

161 Guide to the Selection and Assessment of Silencers.

2.2 Indian Standard

IS 3483 : 1965 Code of practice for noise reduction in industrial buildings

IS 12065 : 1987 Permissible limits of noise level for rotating electrical machines

IS 9989 : 1981 Assessment of Noise with Respect to Community Response

IS 7194 : 1994 /ISO 1999 : 1990 Assessment of Noise Exposure During Work for Hearing Conservation Purpose .

3.0 GENERAL REQUIREMENTS

3.1. EEMUA Publication 140 forms an integral part of this Specification and definitions given in Publication 140.

3.2. Noise measurements reporting, calculations and acceptance testing on equipment shall be made in accordance with EEMUA Publication 140. Alternative, equivalent, national test procedures may be used subject to written approval from the IOCL.

3.3. Where large equipment, such as a fired heater, has to be treated as a number of noise emitting components, the vendor shall provide information for each of these separate components, obtained in accordance with the procedures of EEMUA Publication 140.

3.4. Noise control measures shall not conflict with the other requirements of the equipment material requisition.

3.5. Silencers and acoustic enclosures forming part of the supply shall comply with EEMUA Publication 106.

4.0 APPLICATION OF NOISE LIMITS

4.1 The equipment noise limits have been selected so that the total plant noise generated shall not exceed the plant limits.

4.2 Sound pressure level limits and applicable locations as stated in the equipment requisition shall apply. If sound pressure level limits are not stated in the requisition or are incomplete then the requirements as defined in paragraphs 5.1 to 5.4 shall apply.

4.3 Sound power level limits, where applicable, shall be specified by the IOCL in the requisitions for specific equipment and shall apply to the total noise produced by all components forming part of the equipment.

4.4 The equipment must comply with the most restrictive of the specified work area or sound power level criteria.

4.5 The noise limits shall be modified in accordance with Section 5.3 when the equipment emits annoying characteristics such as narrow-band or impulsive noise. The modification for narrow-band noise shall apply only to the octave band or bands in which the characteristics occur.

4.6 If the equipment emits a fluctuating or intermittent noise the equivalent continuous sound level over the most noisy 8 hours shall meet the specified limits. The maximum level shall not be more than 10 dB higher than the specified limits.

4.7 Individual octave band limits may be exceeded by up to 3 dB by agreement with the IOCL provided the overall dB(A) limit is met.

5.0 NOISE LIMITS

The noise generated by the equipment shall not exceed the noise limits given in the following sections, for any of the conditions of operation, including turndown, specified in the equipment material requisition and during start-up, shut-down and commissioning of the equipment. The limits apply for the equipment as installed on site but excluding any reverberation effects.

Sound pressure level limits apply at 1 m from the equipment surface unless otherwise stated in Section 5.4.

Restricted limits may be permitted for certain defined equipment. Such equipment will be located in restricted access areas.

5.1 Sound Pressure Level Limits (dB re 2 x 10^{-5}N/m^2)

Limit Type	Work Area Limit Code	dBA	OCTAVE-BAND CENTRE FREQUENCY (Hz)								
			31.5	63	125	250	500	1k	2k	4k	8k
General	G1	85	109	98	91	86	83	80	78	76	75
	G2	82	107	96	89	83	80	77	75	73	72
	G3	79	106	94	86	80	77	74	72	70	68
	G4	75	103	91	83	77	73	70	68	66	64
	G5	70	99	87	79	72	68	65	63	61	59
Restricted	R1	90	117	103	96	91	88	85	83	81	80
	R2	120									

5.2 Sound Power Level Limits (dB re 10^{-12}W)

Section 5.4 below defines those equipment items for which sound power level limits may apply in addition to the limits of Section 5.1 above.

Sound power level limits, where applicable, are specified by the IOCL in the material requisition for the specific equipment and shall apply to the total noise produced by all components forming part of the equipment.

5.3 Reduction in Noise Limits for Narrow Band or Impulsive Noise

Characteristic	dBA	REDUCTION OF LIMIT IN SECTION 5.2 (dB) OCTAVE-BAND CENTRE FREQUENCY (Hz)							
		63	125	250	500	1k	2k	4k	8k
Narrow Band Noise	-	0	0	0	4	6	7	7	3
Impulsive Noise	5	-	-	-	-	-	-	-	-

5.4 Applicable Noise Limits for Equipment

Equipment	Work Area Limit Code	Applicable Locations	Sound Power Level Limit
Pumps, Gears, Blowers, Agitators/Mixers and their Drivers	G3 G2	1m from individual equip. item 1m from equip. for complete train	- -
Centrifugal and Axial Compressors, Drivers and Gears over 1000 kW. Excluding Gas Turbines	R1 G5	1m from equip. train Inside Operator refuge, if provided	See Requisition
Centrifugal Compressors up to 1000 kW, Other compressors, Compressor Drivers and Gears	G1 G2	1m from equip. train 1m from individual equip. item	- -
Fired Heaters and Boilers including Fans and Drivers all operating	G1 G1 G1	1m from burners and below floor 1m from equip. surface excluding burners and area below floor 1m from furnace perimeter 1.5m above grade	See Requisition
Air Cooled Exchangers all running	G1 G3	1m below fans or tube bundle 1.5m above grade below unit	See Requisition
Cooling Tower Operating with all Fans running	G1 G2	1m from fan stack on fan deck 1m from air intake	See Requisition
Gas Turbines and Large Equipment fitted with Acoustic Enclosure	G1 R2 G5	1m from acoustic enclosure Inside enclosure for equip. Inside operator refuge, if provided	See Requisition
Operating Vents	G1	At 3m from vent or on nearest platform	-
Emergency Vents	R2	At 3m from vent or on nearest platform	-
Valves and Piping Components	G2	1m from equip. and piping	-
Other Noise Sources Located Outdoors	G2	1m from equipment	See Requisition

Noise Sources Located Inside Process Plant Buildings	G3 G4	1m from individual widely spaced items 1m from individual equip. closely spaced	- -
Flare	G1 G1	1m from equipment at grade 50m horizontally from flare stack at grade	See Requisition

6.0 INFORMATION TO BE SUBMITTED WITH THE QUOTATION

The Vendor shall state noise guarantees for the equipment he offers to supply. The Equipment Noise Data Sheet or Noise Data Tables forming part of the equipment Material Requisition, shall be returned with the quotation completed by the Vendor. Additional information shall be provided on separate sheets when required.

If the Noise Data Sheet has not been included with the equipment Material Requisition, then the Vendor shall submit his noise data on a copy of the typical Noise Data Sheet attached to this standard.

Information supplied by the Vendor with his quotation shall include the following:

a) Guaranteed sound pressure levels in dBA and octave bands (31 Hz - 8KHz) at 1m from the equipment surface for equipment of standard design and of special design or with noise control features as appropriate.

b) If applicable, guaranteed sound power levels in dBA and octave bands (31 Hz - 8KHz) for equipment of standard design and of special design or with noise control features. The calculation method shall be shown in detail. Where equipment sound pressure levels guaranteed by Vendor exceed 85 dBA, sound power levels shall also be supplied even if not specifically required by the Requisition.

c) Expected or measured noise levels in dBA and octave bands (31 Hz - 8KHz) for equipment of standard design and of special design or with noise treatment.

d) The Vendor shall state the operating conditions for his equipment at which noise levels emitted by the equipment are at a maximum (normally full load). Guaranteed noise levels will apply at sated conditions unless otherwise agreed by the IOCL.

e) The Vendor shall state whether the noise source exhibits distinctive narrow band or impulsive characteristics and indicate the dominant frequencies.

f) Where predictions of noise levels are made or where information is extrapolated from test data, then the basis of the prediction and extrapolation methods shall be given.

g) Details of noise-reducing features included in the supply and information on experience with these features.

h) If silencing measures outside the Vendors scope are necessary to meet the specified noise limits, the Vendors shall state any such requirements.

i) Wherever possible noise test reports should be submitted, since this may enable the IOCL to waive the requirements for any noise acceptance testing. In any case additional costs associated with noise acceptance testing should be stated.

7.0 NOISE ACCEPTANCE TESTING

7.1 When specified in the equipment requisition, noise tests will be required to be undertaken as part of the general acceptance tests at the successful Vendor's works, or other suitable location.

7.2 Noise tests shall be conducted in accordance with Part 2 of EEMUA 140 or equivalent national or international procedure approved in writing by the IOCL.

7.3 A written test report shall be submitted to the IOCL within two weeks of completion of the test and shall contain, as a minimum, all information required by EEMUA 140.

7.4 In additional to any equipment noise tests by the Vendor, the IOCL or their agents will carry out noise measurements on equipment as part of the acceptance testing for the complete plant.

7.5 Should the results of Vendor's shop tests or subsequent field test show that guaranteed noise levels are not being met, the Vendor shall be responsible for any necessary modifications to his equipment agreed with the IOCL. Such modifications shall be carried out at times designated by the IOCL.

ATTACHMENT - EQUIPMENT NOISE DATA SHEET

EQUIPMENT DESCRIPTION:

MAXIMUM ALLOWABLE NOISE LEVELS: (Without impulse or tonal component)

Limit Type	Location for Lp or Component for Lw	Octave Band Centre Frequency (Hz) 31.5 63 125 250 500 1K 2K 4K 8K	dBA
Sound Pressure			
Sound Power Level (Lw)			

VENDORS NOISE LEVEL DATA

State Operating conditions for maximum noise and guarantee:

X - Guaranteed Levels (Upper tolerance +0db)

Limit Type		Location for Lp or Component for Lw	Octave Band Centre Frequency (Hz) 31.5 63 125 250 500 1K 2K 4K 8K	dBA
Sound Pressure Level(Lp)	Standard Equipment			
	Special Design			
Sound Power Level(Lw)	Standard Equipment			
	Special Design			

X - Expected/Measured Noise Levels

Limit Type	Location for Lp or Component for Lw	Octave Band Centre Frequency (Hz) 31.5 63 125 250 500 1K 2K 4K 8K	dBA
Standard Equipment			
Special design or Noise control			

Noise level Obtained from:

TEST: Equipment identical to offered equipment
Extrapolation of data for similar equipment
Other method (give details)

Noise characteristics
Narrow band YES/NO
Impulsive YES/NO

Other YES/NO Description

Noise test Report attached YES/NO
VENDOR'S SIGNATURE:_____ DATE:

NOTES: Supporting evidence (noise test reports etc) for vendor's quoted noise levels should be enclosed.

X - Items to be completed by vendor
Lp - Sound pressure level dB re $2*10^{-5}$ Pa at 1m from equipment surface (unless otherwise specified).
Lw - Sound power level dB re 10^{-12}W